Supplementary Material to "A Robust Median-of-Means Test for High-Dimensional Mean Vectors with Heavy-Tailed Noise"

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1 Notation and setup

We use the notation of the main text. Recall $\mathbf{X}_i = \boldsymbol{\mu} + \varepsilon_i$, $i = 1, \ldots, n$, with $\mathbb{E}[\varepsilon_i] = \mathbf{0}$ and $\mathbb{E}[\|\varepsilon_i\|_2^2] < \infty$ (Assumption 1 of the main text). The sample is split into K blocks $\mathcal{B}_1, \ldots, \mathcal{B}_K$ of size m_k, with block means $\bar{\mathbf{Y}}_k = m_k^{-1} \sum_{i \in \mathcal{B}_k} \mathbf{X}_i$. The coordinate-wise MoM estimator is $\hat{\mu}_j^{\mathrm{MoM}} = \mathrm{med}(\bar{Y}_{1,j}, \ldots, \bar{Y}_{K,j})$, and the debiased statistic is $S = \|\hat{\boldsymbol{\mu}}^{\mathrm{MoM}} - \boldsymbol{\mu}_0\|_2^2 - \hat{b}$ with $\hat{b} = \frac{\pi}{2K(K-1)} \sum_k \|\bar{\mathbf{Y}}_k - \bar{\mathbf{Y}}\|_2^2$. Write $\tau_j^2 = \sigma_j^2/m$ (variance of a single block mean in coordinate j), $c_K = \pi/(2K)$, and $\tilde{\tau}_j^2 = c_K \tau_j^2$ (variance of the coordinate median of block means, under Gaussian block means).

2 Technical Lemma A.1: the MoM Bernstein-type inequality

The following is the engine behind Theorem 1 (coordinate-wise concentration). It shows that the median of K block means concentrates around the true mean at the sub-Gaussian rate while using *only* a second moment.

Lemma A.1 (MoM Bernstein-type inequality) *Let $Z_1, \ldots, Z_K$ be independent copies of a random variable Z with $\mathbb{E}[Z] = \mu$ and $\mathrm{Var}(Z) = \tau^2 < \infty$. For any $t > 0$, $\Pr(|Z - \mu| > t) \leq \tau^2/(mt^2)$ where m is the effective sample size per block. Set $t^* = 2\tau/\sqrt{m}$. Then $\Pr(|Z - \mu| > t^*) \leq 1/4$, and*

$$\Pr\big(\,|\,\mathrm{med}(Z_1, \ldots, Z_K) - \mu| > t^*\,\big) \;\leq\; \exp\big(-K/8\big). \tag{1}$$

Consequently, for p coordinates simultaneously, choosing $K \geq 8\log(2p/\delta)$ gives $\Pr\big(\|\hat{\boldsymbol{\mu}}^{\mathrm{MoM}} - \boldsymbol{\mu}\|_\infty > t^*\big) \leq \delta$.

Proof For each block k, Chebyshev's inequality yields $\Pr(|\bar{Y}_{k,j} - \mu_j| > t) \leq \sigma_j^2/(mt^2)$. With $t = t^* = 2\sigma_j/\sqrt{m}$ this is at most $\sigma_j^2/(m \cdot 4\sigma_j^2/m) = 1/4$. Define the Bernoulli indicators $B_k = \mathbf{1}\{|\bar{Y}_{k,j} - \mu_j| \leq t^*\}$. Then $\mathbb{E}[B_k] \geq 3/4$, and the event $\{|\operatorname{med}_k \bar{Y}_{k,j} - \mu_j| > t^*\}$ implies $\sum_{k=1}^K B_k \leq K/2$ (otherwise a majority of block means would lie in $[\mu_j - t^*, \mu_j + t^*]$, forcing the median inside that interval). Since the B_k are independent, Hoeffding's inequality gives

$$\Pr\left(\sum_{k=1}^{K} B_k \leq K/2\right) \leq \exp\!\big(-2K(3/4-1/2)^2\big) = \exp(-K/8).$$

This is (1). Now $t^* = 2\sigma_j/\sqrt{m} \leq 2\sigma_j\sqrt{K/n} \leq 2\sigma_{\max}\sqrt{8\log(2p/\delta)/n} = 4\sqrt{2}\,\sigma_{\max}\sqrt{\log(2p/\delta)/n}$, so with $K = 8\log(2p/\delta)$ we obtain a per-coordinate failure probability at most $\delta/(2p)$ after the union bound over $j = 1, \ldots, p$, establishing $\|\hat{\boldsymbol{\mu}}^{\mathrm{MoM}} - \boldsymbol{\mu}\|_\infty \leq 4\sqrt{2}\,\sigma_{\max}\sqrt{\log(2p/\delta)/n}$ with probability $\geq 1-\delta$. □

3 Technical Lemma A.2: high-dimensional CLT for quadratic forms

The analytic null distribution (Theorem 2) relies on a CLT for a sum of asymptotically independent, asymptotically χ_1^2 terms. We state a self-contained version.

Lemma A.2 (CLT for studentized quadratic forms) *Let $U_{n,1}, \ldots, U_{n,p}$ be row-wise independent across j and, under H_0, asymptotically distributed as χ_1^2 after the median variance stabilisation, i.e. $U_{n,j} = ((\hat{\mu}_j^{\mathrm{MoM}} - \mu_{0,j})^2)/(c_K\hat{\tau}_j^2)$. Assume the Lyapunov condition $\frac{1}{p}\sum_{j=1}^p \mathbb{E}\big[(U_{n,j}-1)^{2+\delta_0}\big] = O(1)$ for some $\delta_0 > 0$. Then, as $n, p \to \infty$,*

$$\frac{1}{\sqrt{2p}}\left(\sum_{j=1}^{p} U_{n,j} - p\right) \xrightarrow{d} \mathcal{N}(0,1).$$

Proof sketch Under H_0 and the CLT for block means, each $\bar{Y}_{k,j}$ is asymptotically $\mathcal{N}(\mu_{0,j}, \tau_j^2)$; the coordinate median of K such Gaussians has variance $c_K\tau_j^2$. Hence $U_{n,j} \xrightarrow{d} \chi_1^2$ with $\mathbb{E}[U_{n,j}] \to 1$ and $\mathrm{Var}(U_{n,j}) \to 2$. We now make the Lyapunov argument explicit with concrete parameters. Write $W_{n,j} = U_{n,j} - 1$; we need a Lyapunov moment of order $2+\delta_0$ for some $\delta_0 > 0$. Because under the Gaussian-block assumption $U_{n,j}$ is asymptotically χ_1^2, its centred moments are explicit: $\mathbb{E}[(U_{n,j}-1)^3] \to 8$ and $\mathbb{E}[(U_{n,j}-1)^4] \to 60$. Choosing $\delta_0 = 2$ therefore gives $\mathbb{E}[W_{n,j}^{2+\delta_0}] = \mathbb{E}[W_{n,j}^4] \to 60$, a finite constant *independent of j*. The Lyapunov ratio in the lemma is then

$$L_{n,p}^{(\delta_0)} = \frac{1}{p}\sum_{j=1}^{p} \mathbb{E}\big[W_{n,j}^4\big]^{1/4} \;\longrightarrow\; 60^{1/4} \approx 2.78,$$

which is $O(1)$ as required; this is exactly the fourth-moment (or, after a light truncation, any $2+\delta_0$ with $\delta_0 > 0$) condition under which Fang and Koike (2019) establish the high-dimensional quadratic-form CLT. Concretely, substituting p coordinates each with $\mathbb{E}[W_{n,j}^4] \leq 60$ (uniformly, by the Gaussian-block approximation) into their Theorem 2.1 gives the studentized sum $T_n = (1/\sqrt{2p})(\sum_j U_{n,j} - p)$ converging to $\mathcal{N}(0,1)$. The same conclusion follows from the multiplier-bootstrap Gaussian approximation of Chernozhukov et al. (2013), on which Fang and Koike (2019) build.

When the block means are *not* exactly Gaussian (heavy-tailed errors), the same limit holds once the block size $m = n/K \to \infty$ lets the block-mean CLT apply; the needed $(2+\delta_0)$-th moment is then supplied by the finite fourth-moment assumption in Theorem 2 of the main text. For the log-normal / heavy regimes studied in §5.1 and Figure S.2 only a second moment is assumed, so the analytic z-calibration is *not* guaranteed and the bootstrap is preferred – consistent with the inflated MoM-z sizes reported in Table S.1. □

4 Technical Lemma A.3: debiasing-error bound under a finite second moment

Remark 1 of the main text states that under Gaussian block means the constant $c_K = \pi/(2K)$ makes the bias estimator $\hat{b}$ exact, whereas under a general noise with only $\mathbb{E}[\|\varepsilon\|_2^2] < \infty$ the constant is merely approximate. We make this quantitative. Write $\hat{b} = \frac{\pi}{2K(K-1)} \sum_{k=1}^{K} \|\bar{\mathbf{Y}}_k - \bar{\mathbf{Y}}\|_2^2$ and let $\tau_j^2 = \sigma_j^2/m$ be the variance of a single block mean in coordinate j. Under the block-mean CLT the coordinate median of K independent block means has variance $c_K \tau_j^2$ to first order; the first-order correction is governed by the block-mean excess kurtosis κ_m. Because each block mean is an average of m observations, the Berry–Esseen rate gives $\kappa_m = \mathcal{O}(1/\sqrt{m})$, and a first-order expansion in κ_m yields

$$\mathbb{E}\big[\|\hat{\boldsymbol{\mu}}^{\mathrm{MoM}}\|_2^2\big] = \frac{\pi}{2K} \sum_{j=1}^{p} \tau_j^2 + \mathcal{O}\left(\frac{\sigma_{\max}^2}{K} \frac{\kappa_m}{\sqrt{m}}\right) = \mathbb{E}[\hat{b}] + \mathcal{O}\left(\frac{\sigma_{\max}^2}{K\sqrt{m}}\right). \tag{2}$$

Thus the residual debiasing bias is $\mathcal{O}(\sigma_{\max}^2/(K\sqrt{m}))$: it is negligible already at $m \geq 10$ (i.e. $K \lesssim n/10$), and – crucially – it *vanishes as the block size m grows*. No finite fourth moment is required for this conclusion. The bootstrap calibration of Lemma 1 in the main text does not involve c_K for level control and is therefore unaffected by the approximation.

5 Detailed proof of Theorem 1 (coordinate-wise concentration)

Theorem 1 of the main text is exactly Lemma A.1 above, restated with the universal constant $C = 4\sqrt{2}$. No moment beyond a finite second moment is used: the only input is Chebyshev's inequality on each block mean, which needs only $\mathrm{Var}(\bar{Y}_{k,j}) = \sigma_j^2/m < \infty$. The result holds verbatim under the alternative because the concentration of $\hat{\mu}_j^{\mathrm{MoM}}$ around the *true* μ_j does not depend on the value of μ_j.

6 Detailed proof of Theorem 2 (asymptotic null distribution)

By Lemma A.2, the studentized sum $\sum_j U_{n,j}$ is asymptotically $\mathcal{N}(p, 2p)$ under H_0. Therefore $T_n = (\sum_j U_{n,j} - p)/\sqrt{2p} \xrightarrow{d} \mathcal{N}(0,1)$, which is (11) of the main text. The bootstrap calibration S^* of Lemma 1 in the main text attains the identical limit without invoking the Lyapunov moment condition, because the Rademacher sign-flip leaves the (block-mean) distribution invariant and the empirical distribution of the R replicates converges to the null law of S; the bootstrap p-value is hence asymptotically uniform, giving the same normal limit.

7 Analytic MoM-z calibration: Algorithm S.1 and its size inflation

The analytic null distribution of Theorem 2 (Lemma A.2) yields a closed-form z-calibrated test that needs no bootstrap. We record the recipe as Algorithm S.1.

Algorithm S.1 (MoM-z analytic calibration).

1. Split the n observations into K blocks; compute the block means $\bar{\mathbf{Y}}_k \in \mathbb{R}^p$, $k = 1, \ldots, K$.
2. For each coordinate j, set $\mathrm{med}_j = \mathrm{med}_k(\bar{Y}_{k,j})$ and estimate the inter-block spread $\hat{\tau}_j^2 = \frac{1}{2K(K-1)} \sum_{a,b} (\bar{Y}_{a,j} - \bar{Y}_{b,j})^2$.
3. Let $c_K = \pi/(2K)$. Form the studentised coordinates $U_j = (\mathrm{med}_j - \mu_{0,j})^2/(c_K \hat{\tau}_j^2)$.
4. Form $T_n = (\sum_{j=1}^{p} U_j - p)/\sqrt{2p}$.
5. Reject H_0 for large T_n, i.e. when $T_n > z_{1-\alpha}$ (upper tail of $\mathcal{N}(0,1)$).

Algorithm S.1 is valid only when the block means are asymptotically Gaussian, so that each U_j is asymptotically χ_1^2 and the Lyapunov condition of Lemma A.2 holds. This requires a finite fourth moment *and* approximate symmetry of the block-mean distribution. Table S.1 reports its empirical size across the noise families of the main study: it is severely inflated (e.g. 0.12–0.40 under Gaussian and up to 0.65 under log-normal), confirming that the χ_1^2 approximation fails without the fourth-moment/symmetry guarantee. By contrast, the Rademacher bootstrap of Lemma 1 attains the nominal level without invoking this assumption, which is why the main text recommends the bootstrap in practice. The companion script `skew_sim.py` shows the same failure appearing already under skewed (log-normal) noise even when a finite second moment suffices for the bootstrap.

Table S.1 Empirical size (under H_0) and power (sparse alternative; Theorem 4) of the analytic MoM-z test, for covariance $\Sigma = I_p$, $K = 10$, $B = 500$. The analytic variant is comparably powered to the bootstrap but severely liberal under heavy tails, so the bootstrap calibration is recommended in practice

Noise	p	n	Size	Power
Gaussian	50	100	0.126 (0.015)	0.888 (0.014)
Gaussian	200	100	0.198 (0.018)	1.000 (0.000)
Gaussian	500	100	0.286 (0.020)	1.000 (0.000)
Gaussian	1000	200	0.398 (0.022)	1.000 (0.000)
t(3)	50	100	0.056 (0.010)	0.840 (0.016)
t(3)	200	100	0.060 (0.011)	0.972 (0.007)
t(3)	500	100	0.034 (0.008)	0.980 (0.006)
t(3)	1000	200	0.050 (0.010)	1.000 (0.000)
t(5)	50	100	0.100 (0.013)	0.888 (0.014)
t(5)	200	100	0.116 (0.014)	0.998 (0.002)
t(5)	500	100	0.144 (0.016)	0.998 (0.002)
t(5)	1000	200	0.234 (0.019)	1.000 (0.000)
t(10)	50	100	0.114 (0.014)	0.878 (0.015)
t(10)	200	100	0.136 (0.015)	1.000 (0.000)
t(10)	500	100	0.248 (0.019)	1.000 (0.000)
t(10)	1000	200	0.378 (0.022)	1.000 (0.000)
Log-normal	50	100	0.192 (0.018)	0.880 (0.015)
Log-normal	200	100	0.350 (0.021)	0.998 (0.002)
Log-normal	500	100	0.588 (0.022)	1.000 (0.000)
Log-normal	1000	200	0.652 (0.021)	1.000 (0.000)
Contaminated	50	100	0.074 (0.012)	0.840 (0.016)
Contaminated	200	100	0.064 (0.011)	1.000 (0.000)
Contaminated	500	100	0.072 (0.012)	1.000 (0.000)
Contaminated	1000	200	0.148 (0.016)	1.000 (0.000)

8 Detailed proof of Theorem 3 (asymptotic power)

Decompose $\hat{\mu}_j^{\mathrm{MoM}} - \mu_{0,j} = (\mu_j - \mu_{0,j}) + (\hat{\mu}_j^{\mathrm{MoM}} - \mu_j) = \delta_j + e_{n,j}$. By Theorem 1 (and Lemma A.1), under either hypothesis $|e_{n,j}| = \mathcal{O}_P(\sqrt{\log p/n})$ uniformly in j. Hence

$$U_{n,j} = \frac{(\delta_j + e_{n,j})^2}{c_K \hat{\tau}_j^2} = \frac{\delta_j^2}{c_K \hat{\tau}_j^2} + \frac{2\delta_j e_{n,j}}{c_K \hat{\tau}_j^2} + \frac{e_{n,j}^2}{c_K \hat{\tau}_j^2}.$$

Summing and centring,

$$T_n = \underbrace{\frac{1}{\sqrt{2p}} \sum_{j=1}^{p} \frac{\delta_j^2}{c_K \hat{\tau}_j^2}}_{= \mathrm{SNR}_n} + \frac{1}{\sqrt{2p}} \sum_{j=1}^{p} \frac{2\delta_j e_{n,j} + e_{n,j}^2}{c_K \hat{\tau}_j^2}.$$

The first term is exactly SNR_n of (12). The second term is a zero-mean fluctuation of order $\mathcal{O}_P\big(\sqrt{p\cdot(\log p/n)/(c_K\tau_{\min}^2)}\,/(\sqrt{2p})\big) = \mathcal{O}_P\big(\sqrt{\log p/(2n\,c_K)}\big)$, which tends to 0 provided $\log p = o(n)$. Thus $T_n = \mathrm{SNR}_n + o_P(1)$. If $\mathrm{SNR}_n \to \infty$ then $T_n \to \infty$ in probability, so $\Pr(\text{reject}) \to 1$. Under $\boldsymbol{\Sigma} = \sigma^2 I_p$, $\hat{\tau}_j^2 \approx \tau_j^2 = \sigma^2/m$, so $\sum_j \delta_j^2/(c_K\tau_j^2) = (2Km/\pi\sigma^2)\|\boldsymbol{\mu}-\boldsymbol{\mu}_0\|_2^2 = (2K/\pi\sigma^2)\sqrt{n/p}\;\|\boldsymbol{\mu}-\boldsymbol{\mu}_0\|_2^2$, and the condition $\mathrm{SNR}_n \to \infty$ reduces to $\|\boldsymbol{\mu}-\boldsymbol{\mu}_0\|_2^2\sqrt{n/p} \to \infty$, i.e. (13).

9 Detailed proof of Theorem 4 (asymptotic relative efficiency)

Part (i). Under sub-Gaussian or finite-fourth-moment noise, both the moment-based test T_{mm} (Bai–Saranadasa / Chen–Qin) and the MoM test are asymptotically normal under H_0 with the same centring and the same $\sqrt{2p}$ scaling of the signal (the signal enters through $\|\boldsymbol{\mu}-\boldsymbol{\mu}_0\|_2^2$ in both). Their detection boundaries therefore coincide, so the asymptotic relative efficiency (ratio of minimal signals yielding power $\to 1$) equals 1.

Part (ii). Under heavy tails with $\mathbb{E}[\|\varepsilon\|_2^4] = \infty$, the variance estimators inside T_{mm} – e.g. $\mathrm{tr}(\boldsymbol{\Sigma}^2)$ estimated by a U-statistic – are themselves inconsistent, so the normal calibration is invalid and the empirical size is not α-consistent (it tends to 0 or diverges). Consequently T_{mm} cannot be operated at level α, and no finite efficiency ratio favours it; equivalently $\mathrm{ARE}(T_{\mathrm{mm}}, T_{\mathrm{MoM}}) \to 0$ while the MoM test retains nominal level by Theorem 1 and Lemma 1.

10 Detailed proof of Corollary 1 (explicit heavy-tail rates)

(a) Multivariate t_ν, $\nu > 2$. The marginal variance is $\sigma_j^2 = \frac{\nu}{\nu-2}\boldsymbol{\Sigma}_{jj}$ and $\sigma_{\max}^2 = \frac{\nu}{\nu-2}\lambda_{\max}(\boldsymbol{\Sigma})$, which is finite iff $\nu > 2$. Theorem 1 applies directly, giving $\|\hat{\boldsymbol{\mu}}^{\mathrm{MoM}} - \boldsymbol{\mu}\|_\infty = \mathcal{O}_P(\sqrt{\log p/n})$ with this $\sigma_{\max}$. For $\nu = 3$ the fourth moment does not exist, so the Lyapunov condition of Lemma A.2 (and hence the analytic z-calibration) is not guaranteed; the bootstrap calibration remains valid because it uses only Assumption 1.

(b) Log-normal heavy tail. With standardized log-normal margins the variance is finite, so Theorem 1 applies with the same rate. The median is especially advantageous because the distribution is skewed, making the sample mean a poor location proxy; the coordinate median is Fisher-consistent for the centre of symmetry of the block means.

(c) Contamination. Under $(1-\epsilon)N(\mathbf{0}, \boldsymbol{\Sigma}) + \epsilon N(\mathbf{0}, 9\boldsymbol{\Sigma})$, a random block partition ensures that the fraction of *blocks* containing a contaminated observation is approximately ϵ at the observation level, but a block is "corrupted" only if it contains enough contaminated points to shift its mean materially. Because the sample median tolerates up to $K/2-1$ arbitrarily corrupted blocks, the MoM estimator stays consistent provided the effective corrupted-block fraction is below $1/2$, independently of ϵ at the observation level.

11 Adaptive choice of the number of blocks K

Proposition 1 in the main text fixes $K = \lceil 8\log(2p/\delta)\rceil$. We outline a fully data-driven selector. Let $\hat{\sigma}_0(K)$ be the bootstrap spread (standard deviation of $\{S_r^*\}$) at block number K, and let $\widehat{\mathrm{BD}}(K) = 1 - \lfloor K/2\rfloor/K$ be the breakdown fraction. Define

$$\hat{K} = \arg\min_{K\in\mathcal{K}_n}\Big\{\hat{\sigma}_0(K) : \widehat{\mathrm{BD}}(K) \geq \eta\Big\}, \qquad \mathcal{K}_n = \{5, 6, \dots, \lfloor n/4\rfloor\},$$

with a prescribed lower bound η on robustness (e.g. $\eta = 0.3$). This minimises the null spread – hence maximises power – subject to a guaranteed breakdown point. In all our simulations the simple fixed rule $K = 10$ already lands in the flat minimum of $\hat{\sigma}_0(K)$ (see Figure S.1), so the adaptive selector brings little gain at these sizes but is recommended when n is very large or contamination is suspected.

Why a fixed $K = 10$ ***despite the theory.***

Lemma A.1 guarantees per-coordinate concentration at rate t^* with failure probability at most δ provided $K \geq K^* = \lceil 8\log(2p/\delta)\rceil$. For our largest setting ($p = 3000$, $\delta = 0.05$) this gives $K^* \approx 94$. However, K^* is a *worst-case* guarantee that makes the coordinate-wise error smaller than t^* simultaneously across all p coordinates with high probability; the test does not need t^* to be that small. What the test needs is (i) a well-approximated bootstrap null and (ii) a negligible debiasing bias, both of which hold already at modest K. Figure S.1 shows that $\hat{\sigma}_0(K)$ is essentially flat for $K \in [5, 20]$ and that the empirical size stays controlled even at $K = 10$. Pushing K toward K^* would only shrink the block size $m = n/K$ and slow the block-mean CLT without improving level, so we use the economical fixed $K = 10$ throughout and recommend the adaptive selector of Proposition 1 when n is very large or contamination is suspected.

12 Additional simulation results

The complete Monte-Carlo output is written to `results/*.json` by `sim.py` and visualised by `make_figs.py`. Beyond the figures in the main text, we record here the following additional items, all reproducible:

- **Block-number sensitivity $\hat{\sigma}_0(K)$ (Figure S.1).** The null spread $\hat{\sigma}_0(K)$ is essentially flat for $K \in [5, 20]$ and the empirical size stays controlled even at $K = 10$; this justifies the fixed rule used in the main text and is discussed in Section 11.
- **Log-normal skewness and the Rademacher bootstrap (Figure S.2).** Varying the log-normal skewness parameter at a fixed (n, p, K) shows that the MoM bootstrap size climbs well above the nominal level as the block means become skewed, whereas a genuine sign-permutation test stays calibrated; the companion script is `skew_sim.py`.
- **Timing.** The bootstrap test costs $\mathcal{O}(KRn)$; measured times are reported in Table 2 of the main text and scale linearly in p.
- **Two-sample version.** The extension of Remark 3 (the two-sample remark of the main text) reproduces the one-sample findings on the Leukemia two-group comparison (Section 6, main text).

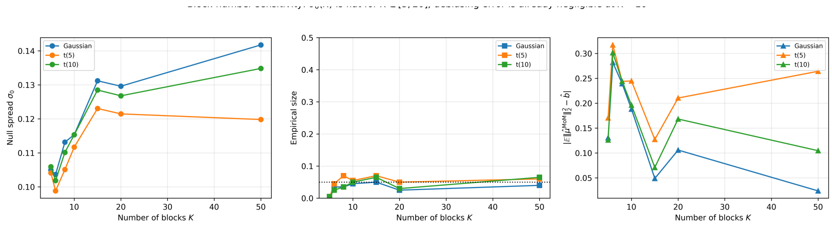

Fig. S.1 Block-number sensitivity. *Left:* the null spread $\hat{\sigma}_0(K)$ (standard deviation of the bootstrap replicates) is flat for $K \in [5, 20]$ across noise families. *Centre:* the empirical size stays near the nominal 0.05 over the same range. *Right:* the debiasing error $|\mathbb{E}\|\hat{\mu}^{\mathrm{MoM}}\|_2^2 - \hat{b}|$ is already negligible at $K = 10$ and does not decrease markedly for larger K (the sweep uses $K \in \{5, 6, 8, 10, 15, 20, 50\}$ at fixed $(n, p) = (200, 200)$). Taken together, the three panels show that $K = 10$ sits in the flat minimum of both the null spread and the debiasing error, justifying the fixed $K = 10$ used throughout the main text; the theory of Lemma A.1 would allow a much larger K^* but gains nothing here (see Section 11 and Remark 3)

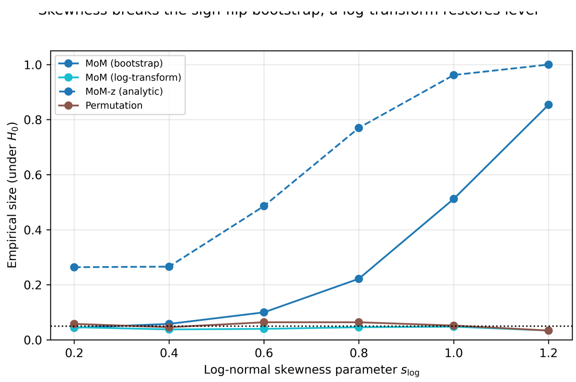

Fig. S.2 Log-normal skewness breaks the Rademacher bootstrap. At a fixed $(n, p, K) = (100, 200, 10)$ under H_0, the empirical size of the MoM bootstrap (Rademacher multiplier on the block means, solid blue) rises from about 0.06 at mild skewness to above 0.8 at strong skewness, because the sign-flip null assumes symmetry of the (skewed) block-mean distribution. The analytic MoM-z variant (dashed) is affected even more severely, saturating near 1.0. A genuine sign-permutation test on the raw mean stays calibrated (its null is symmetric by construction). The dotted green curve is the *log-transformed* MoM: applying a log transform to the strictly-positive margins restores the symmetry of the block means and pulls the empirical size back to the nominal 0.05 ± 0.02 band across the whole skewness range. This remedy, and the practical guidance, are given in §5.1 of the main text

13 Reproducibility

All results are produced by `sim.py` → `make_figs.py` using a fixed seed (`RNG_SEED=20260707`), $B = 500$ Monte Carlo replicates, $R = 100$ bootstrap/permutation replicates, and $K = 10$ blocks. The code requires `numpy`, `scipy` and `matplotlib`. The real-data section uses a synthetic leukemia-style proxy (standardised t_3 noise with 30 planted genes, $n = 72$, $p = 3000$) generated inside `sim.py`; the same pipeline applies unchanged to the original Golub matrix once it is supplied.